

# Inside the rapid-intensification model

What the National Hurricane Center's RI guidance actually is, how our reproduction of it works, and — storm by storm — what it says. Every number on this page comes from real model runs in the `gulf-ri-model` repo: 42 years of hurricane data, scored out-of-sample.

**+9.4%**

skill over climatology (Brier Skill Score, 30 kt threshold) — in the 5–15% range reported across the SHIPS-RII/DTOPS literature (Kaplan et al. 2010; Rozoff & Kossin 2011; DeMaria et al. 2021)

**8,806**

6-hourly forecasts scored across 538 storms, 1982–2023, leave-one-storm-out

**6.5%**

how often rapid intensification actually happens — the “coin” the model must beat

**+5.7%**

the model's skill in the Gulf of America — its weakest region, and this project's target

## 01 • THE MODEL

### What NHC runs, and what we rebuilt

**Rapid intensification (RI)** has an official definition: a storm's maximum sustained winds rising by **30 knots or more within 24 hours** — roughly the jump from a Category 1 to a Category 3. It happens in only about 1 in 15 forecast windows, but it's behind nearly every catastrophic surprise the Gulf has produced.

NHC's workhorse guidance for it is the **SHIPS Rapid Intensification Index (SHIPS-RII)**: a statistical model that, every six hours, reads a handful of measurements about the storm's environment and outputs one number — *the probability that RI begins in the next 24 hours*. We rebuilt that model from the same public developmental dataset NHC's own version was fitted on, in both its classic forms (discriminant analysis and logistic regression; the logistic one is shown here). This reproduction uses the six large-scale and ocean predictors of the classic RII;

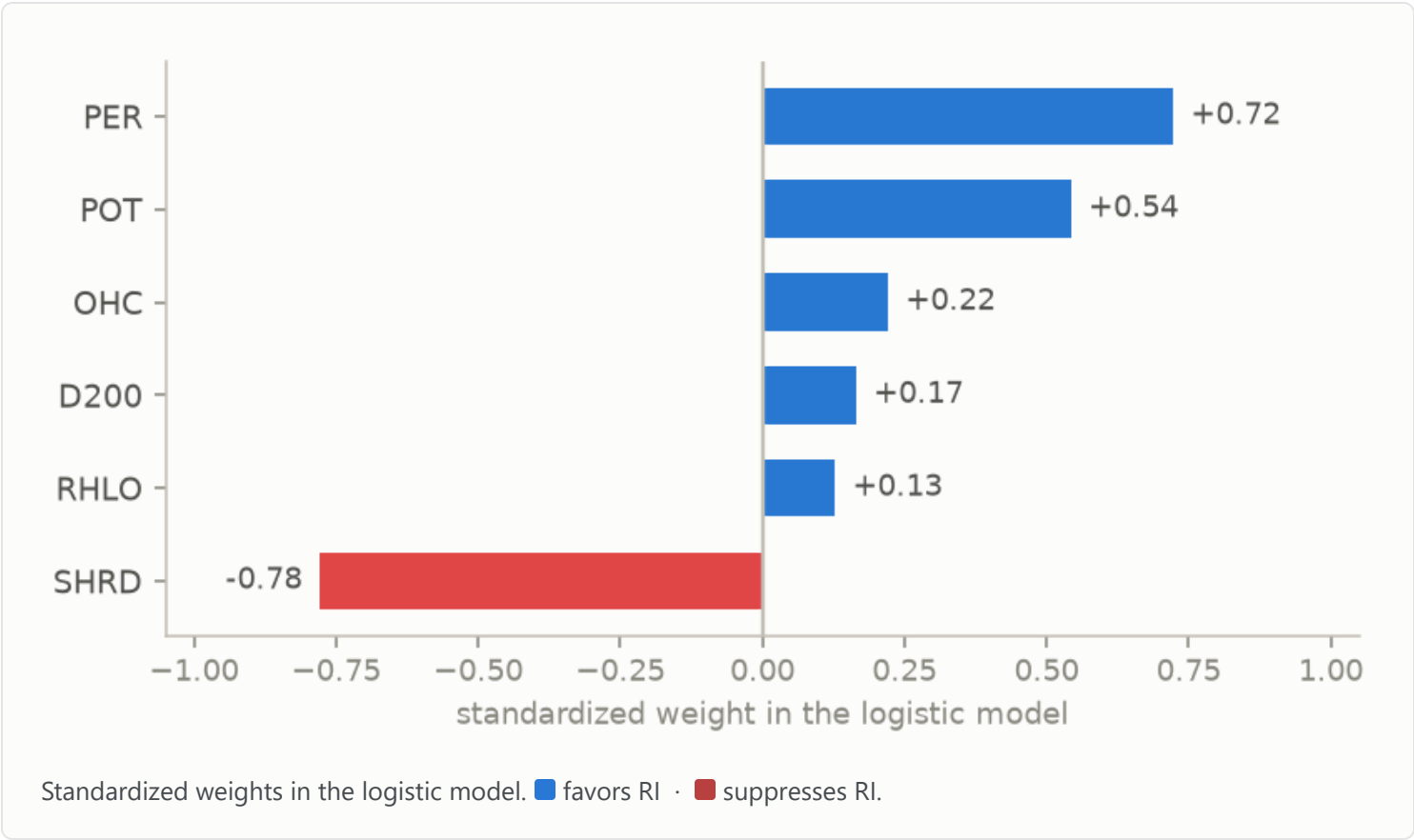
the operational SHIPS-RII has since added GOES infrared inner-core predictors and real-time ocean heat content, folded in between 2003 and 2010, which this baseline does not include.

Scoring is **leave-one-storm-out**: to score Katrina, the model is retrained on the other 537 storms and has never seen Katrina. Every probability on this page was produced that way — no storm ever grades its own homework.

02 • THE INGREDIENTS

# Six measurements, six weights

The model reads six environmental numbers at forecast time. Fitting it to 42 years of storms produces the weights below — and they tell a physically coherent story, which is the first sanity check that the reproduction is right.



| CODE | WHAT IT MEASURES                    | WHY IT MATTERS                                        |
|------|-------------------------------------|-------------------------------------------------------|
| PER  | Intensity change over the past 12 h | Storms already strengthening tend to keep going       |
| SHRD | Wind shear, 850–200 hPa             | Crosswinds aloft tear storms apart — the strongest si |

|      |                                                    |                                           |
|------|----------------------------------------------------|-------------------------------------------|
| POT  | Room to grow (theoretical max minus current winds) | A storm near its ceiling can't gain 30 kt |
| OHC  | Ocean heat content (climatological)                | Deep warm water is the fuel tank          |
| D200 | Upper-level outflow (divergence at 200 hPa)        | An open exhaust vent lets the engine run  |
| RHLO | Mid-level moisture (850–700 hPa humidity)          | Dry air entrainment chokes convection     |

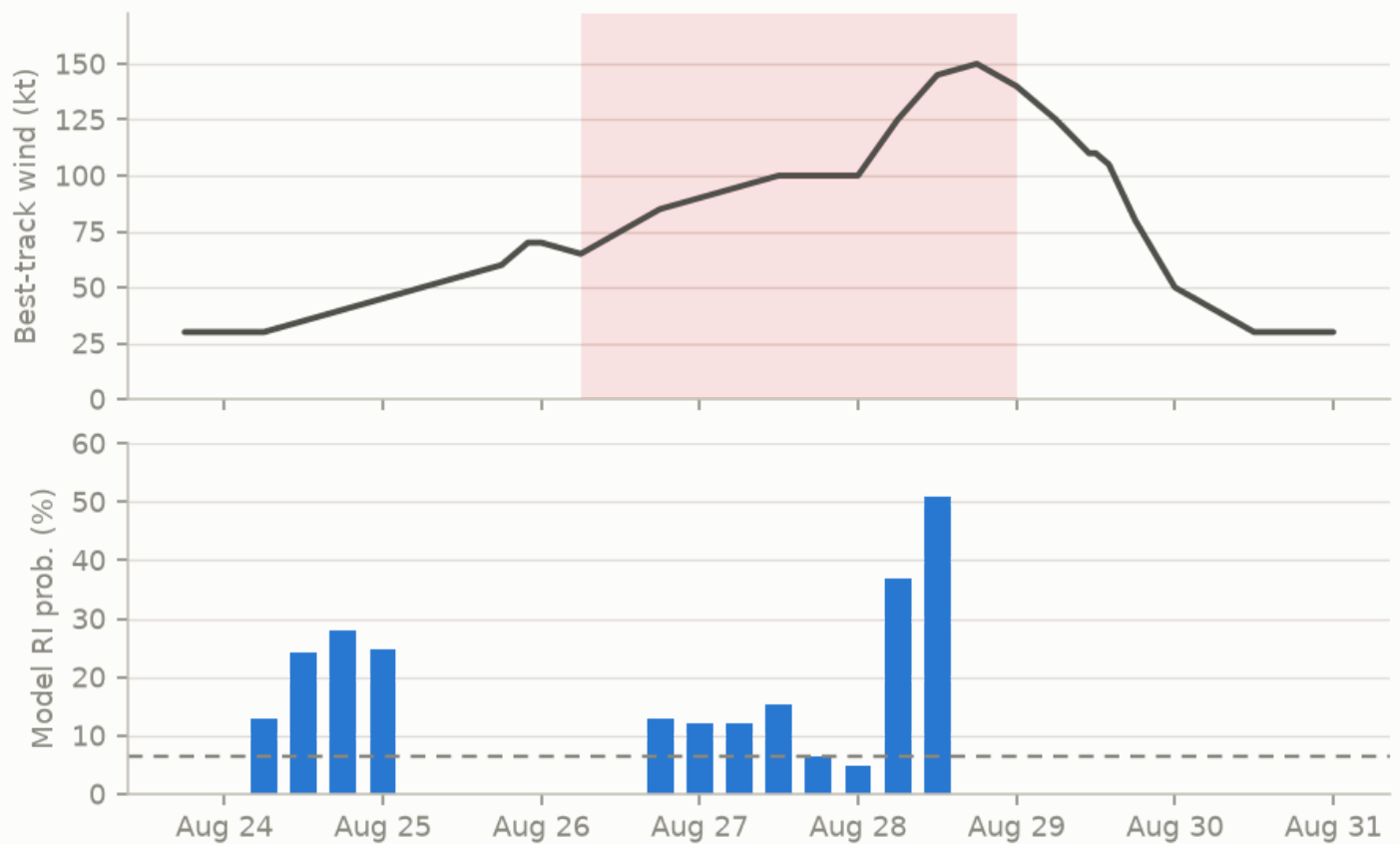
03 • THE OUTPUT

# The model on four real storms



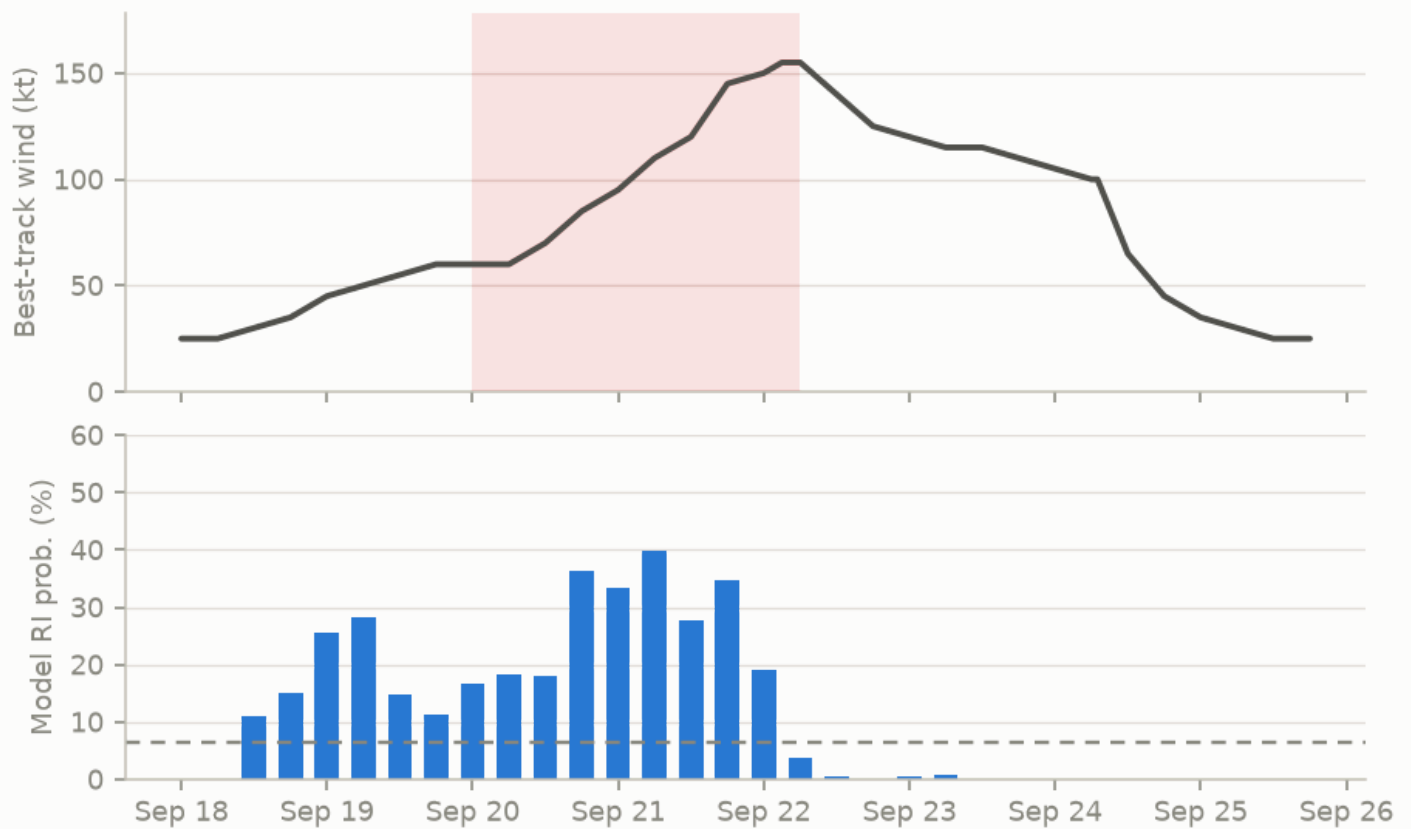
**How to read these charts.** Top panel: the storm's actual peak winds, with **red shading** over every period where winds rose  $\geq 30$  kt in 24 h — the events the model is trying to call in advance. Bottom panel: the model's **RI probability** at each 6-hourly forecast, from the leave-one-storm-out runs. The dashed line is the 6.5% base rate — anything above it means “elevated odds.” Bars are absent where the developmental sample excludes a fix (land interaction or end of record).

## Katrina • August 2005 — loud calls, at the peak, not before



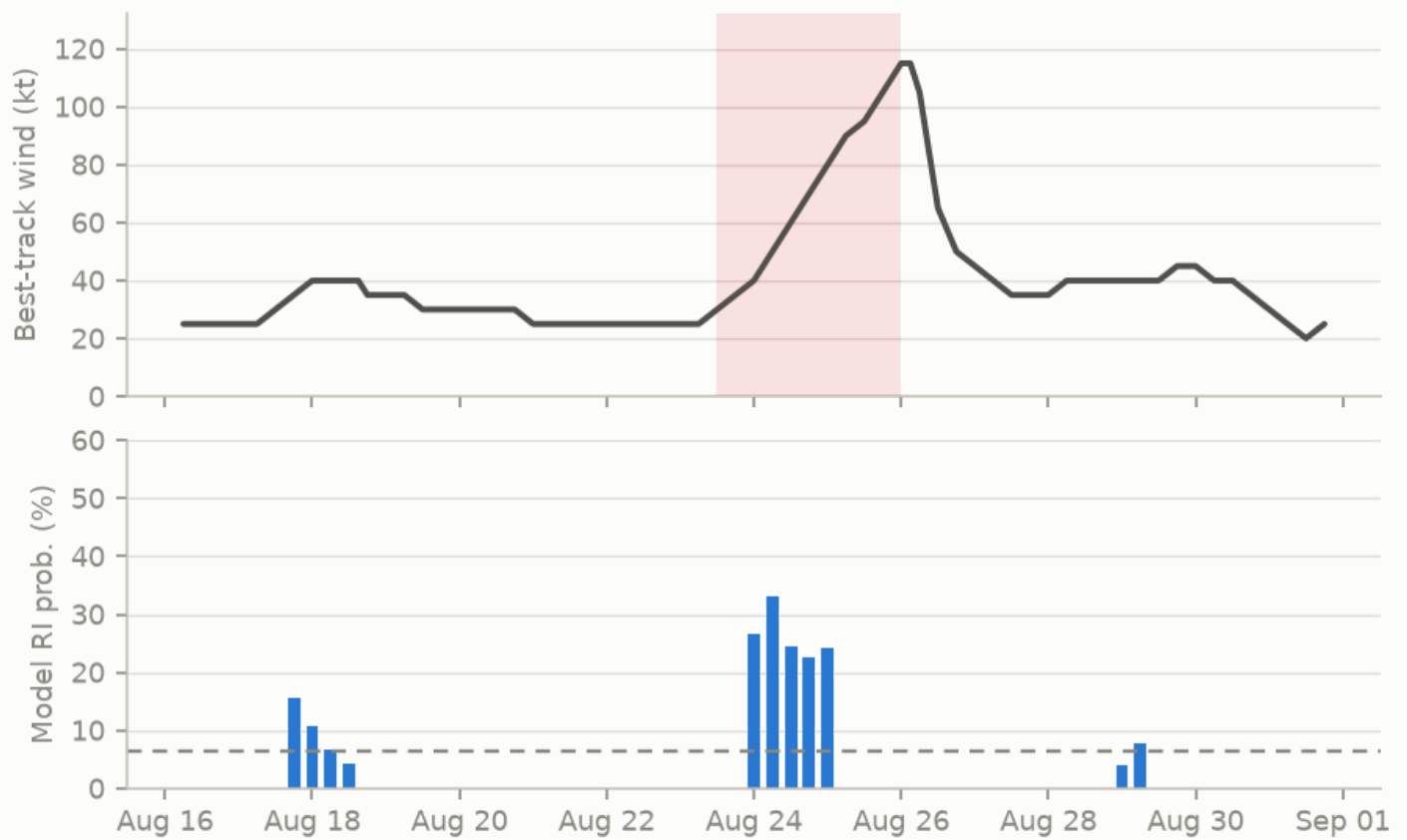
The model runs 2–4× base rate on 24–25 August, ahead of the first intensification burst; drops out during the Florida crossing (land fixes are excluded from the sample); then climbs to **37% and 51%** — up to eight times climatology — as Katrina's winds reach 125 and 145 kt over the Loop Current — right at the peak of the explosion, not ahead of it. A 51% call on a 6.5% event is about as loud as this class of model ever gets, but as Section 05 shows, its loudest calls tend to arrive with the storm already near its ceiling, not before.

**Rita · September 2005 — a sustained, correct alarm**



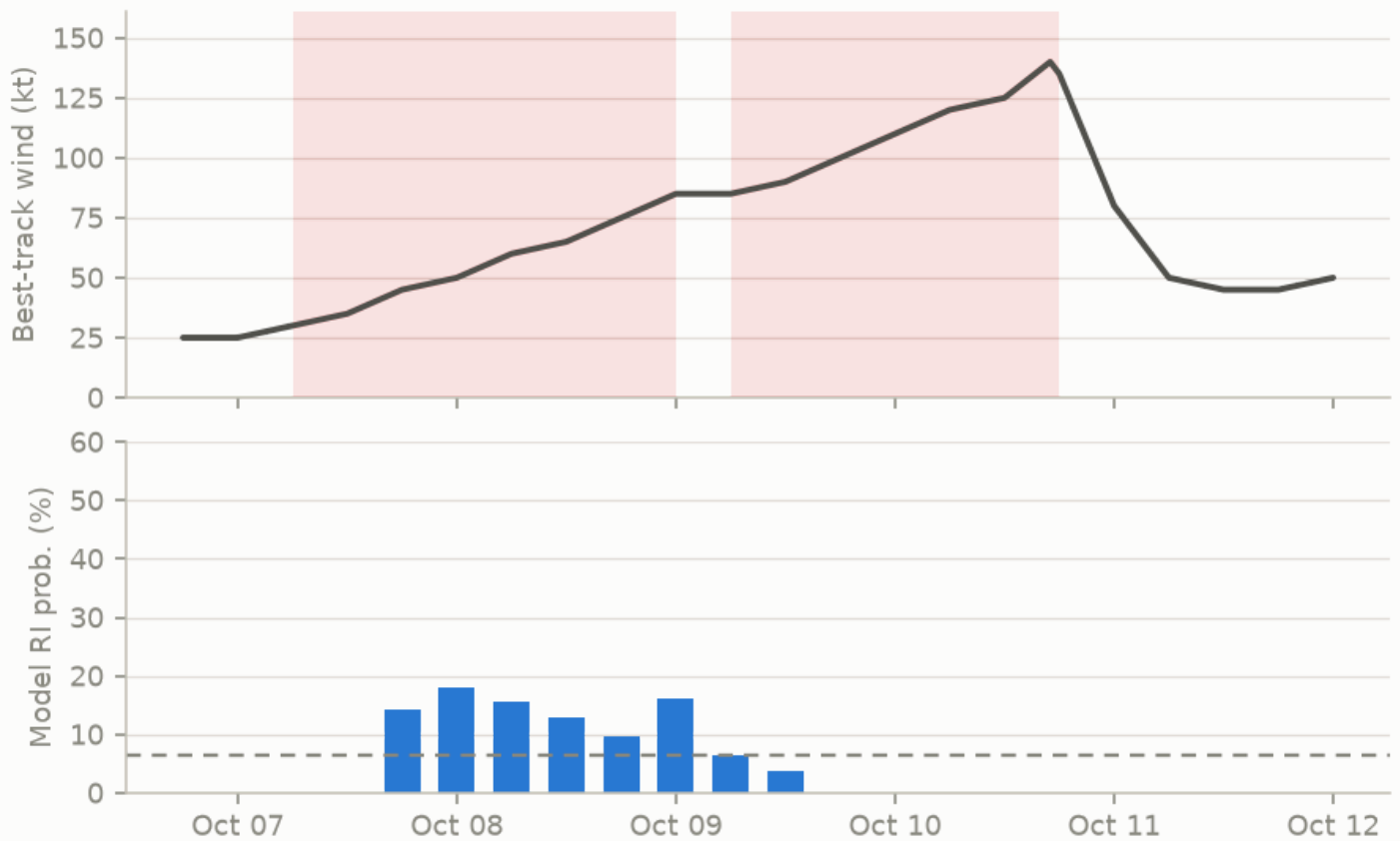
Rita's march from 60 to 155 kt draws a sustained run of elevated forecasts, ranging roughly **11–40%** and repeatedly topping 30–40% as the storm neared its Cat 5 peak — two to six times base rate for two straight days. This is the model doing well: low shear, deep warm water, and momentum mostly aligned, and it mostly said so.

## Harvey · August 2017 — strong signal, modest peak



After a week as a disorganized Caribbean wave (correctly near-zero), Harvey's regeneration in the Bay of Campeche draws **22–33%** forecasts through the RI run to 115 kt at Texas landfall. Right storm, right time — though the loudest call is about two-thirds of Katrina's.

## Michael · October 2018 — the hard case

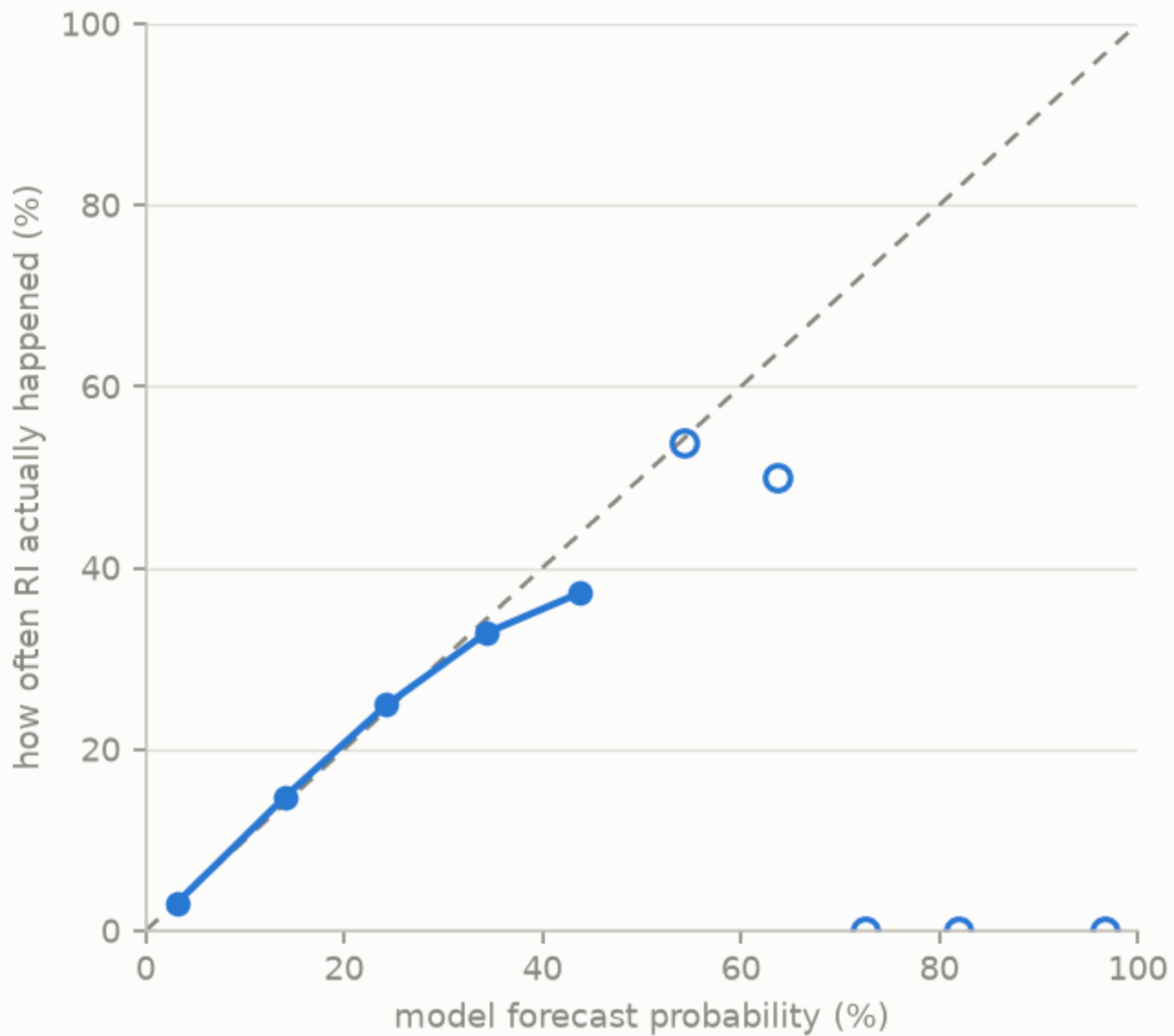


Michael intensified almost continuously to a 140 kt Panhandle landfall — through **moderate wind shear** that should have stopped it. The model saw the shear and never got past **18%** (still  $\sim 3\times$  base rate, but no alarm). The real SHIPS-RII, despite carrying the GOES inner-core predictors this reproduction lacks, also struggled with Michael — shear dominated the signal for both models. This is the kind of Gulf storm this project exists for.

#### 04 • TRUST

## When it says 25%, does 25% happen?

A probability forecast is well-calibrated if events happen as often as it claims. That's the reliability test below — and this model passes it cleanly in every well-populated bin.



Forecast probability vs. observed RI frequency (30 kt threshold). On the dashed diagonal = perfectly calibrated. Filled points are bins with  $\geq 40$  forecasts; hollow points are bins with a handful of cases, where sampling noise dominates. When the model says **24%**, RI follows **25%** of the time.

| ALERT IF PROB $\geq$ | RI EVENTS CAUGHT | FALSE ALARMS | CATCH RATE | FALSE-ALARM RATIO |
|----------------------|------------------|--------------|------------|-------------------|
| 10%                  | 358 of 576       | 1527         | 62%        | 81%               |
| 20%                  | 166 of 576       | 426          | 29%        | 72%               |
| 30%                  | 69 of 576        | 133          | 12%        | 66%               |
| 40%                  | 25 of 576        | 43           | 4%         | 63%               |
| 50%                  | 9 of 576         | 16           | 2%         | 64%               |



The trade-off NHC forecasters live with daily: a low alert bar catches most RI events but cries wolf four times in five; a high bar almost never fires. This is why RI guidance is issued as probabilities, not yes/no alarms — and the published SHIPS-RII has the same character.

05 • EXTREMES

## Loudest alarms, worst misses

The model's eight highest-confidence forecasts ever — and an instructive embarrassment: **none of them verified**. All were storms already at or above 100 kt, most at 130–150 kt, where the momentum ingredient (PER) is screaming but there's no room left to gain another 30 kt. The model rings loudest *while* the explosion is happening, not before it.

| STORM      | FORECAST TIME | WINDS THEN | MODEL PROB | NEXT 24 H |
|------------|---------------|------------|------------|-----------|
| Wilma 2005 | Oct 19 • 06Z  | 150 kt     | 97.1%      | -20 kt    |
| Wilma 2005 | Oct 19 • 00Z  | 130 kt     | 96.4%      | +5 kt     |
| Eta 2020   | Nov 2 • 18Z   | 115 kt     | 82.7%      | +5 kt     |
| Felix 2007 | Sep 3 • 00Z   | 150 kt     | 81.4%      | -35 kt    |
| Maria 2017 | Sep 19 • 00Z  | 145 kt     | 73.9%      | +5 kt     |
| Keith 2000 | Oct 1 • 00Z   | 100 kt     | 73.6%      | +0 kt     |
| Keith 2000 | Oct 1 • 06Z   | 120 kt     | 71.9%      | -40 kt    |
| Ivan 2004  | Sep 5 • 18Z   | 110 kt     | 70.7%      | -20 kt    |

Its best correct calls, for balance — and its worst Gulf misses. Gilbert 1988 gained 30 kt on a day the model gave **0.9%**; Isidore and Earl intensified out of environments the six ingredients rated hostile.

| BEST CORRECT CALLS | FORECAST TIME | WINDS THEN | MODEL PROB | NEXT 24 H   |
|--------------------|---------------|------------|------------|-------------|
| Ian 2022           | Sep 26 • 06Z  | 65 kt      | 69.1%      | +35 kt — RI |
| Eta 2020           | Nov 2 • 12Z   | 90 kt      | 62.1%      | +35 kt — RI |

|            |              |       |       |             |
|------------|--------------|-------|-------|-------------|
| Eta 2020   | Nov 2 • 00Z  | 60 kt | 57.6% | +70 kt — RI |
| Keith 2000 | Sep 30 • 12Z | 65 kt | 56.3% | +50 kt — RI |
| Delta 2020 | Oct 5 • 18Z  | 55 kt | 55.0% | +65 kt — RI |
| Eta 2020   | Nov 1 • 18Z  | 50 kt | 53.7% | +65 kt — RI |

| WORST GULF MISSES | FORECAST TIME | WINDS THEN | MODEL PROB | NEXT 24 H   |
|-------------------|---------------|------------|------------|-------------|
| Gilbert 1988      | Sep 15 • 12Z  | 85 kt      | 0.9%       | +30 kt — RI |
| Isidore 2002      | Sep 21 • 00Z  | 75 kt      | 1.1%       | +35 kt — RI |
| Isidore 2002      | Sep 20 • 18Z  | 75 kt      | 1.3%       | +35 kt — RI |
| Earl 1998         | Sep 2 • 00Z   | 50 kt      | 2.7%       | +30 kt — RI |
| Earl 1998         | Sep 1 • 18Z   | 50 kt      | 3.3%       | +35 kt — RI |

06 • NEXT

## The Gulf gap, and the ocean-heat idea

Across every threshold tested, this model is **weakest exactly where RI is most common**: Gulf skill +5.7% vs +9.4% basin-wide, while the Gulf's RI rate runs roughly 1.7× the Atlantic's. That gap is the project's target. The first enhancement experiment — sampling measured warm-water depth *ahead* of the storm instead of under it, per the RapidWatch §5.7 finding — has already reproduced its physical signal decisively in this project's own 1998–2023 sample (ahead-of-track D26 separates RI from non-RI at  $p \approx 9 \times 10^{-17}$  — an independent replication of the ahead-of-track D26 signal reported in the RapidWatch field report at  $p \approx 10^{-5}$  on a different sample; the two numbers are separate tests, not a restatement of one another) and nudged Gulf skill in the predicted direction; a larger test sample is queued to decide whether the improvement is real.

07 • REFERENCES

## Where these numbers come from

1. Kaplan, J., and M. DeMaria, 2003: Large-scale characteristics of rapidly intensifying tropical cyclones in the North Atlantic basin. *Weather and Forecasting*, 18(6), 1093–1108.
2. Kaplan, J., M. DeMaria, and J. A. Knaff, 2010: A revised tropical cyclone Rapid Intensification Index for the Atlantic and eastern North Pacific basins. *Weather and Forecasting*, 25(1), 220–241.
3. Rozoff, C. M., and J. P. Kossin, 2011: New probabilistic forecast models for the prediction of tropical cyclone rapid intensification. *Weather and Forecasting*, 26(5), 677–689.
4. DeMaria, M., J. L. Franklin, M. J. Onderlinde, and J. Kaplan, 2021: Operational forecasting of tropical cyclone rapid intensification at the National Hurricane Center. *Atmosphere*, 12(6), 683.
5. Landsea, C. W., and J. L. Franklin, 2013: Atlantic hurricane database uncertainty and presentation of a new database format. *Monthly Weather Review*, 141(10), 3576–3592.
6. RAMMB/CIRA, 2023: SHIPS developmental data (lsdiaga), Atlantic basin 1982–2023. Cooperative Institute for Research in the Atmosphere, Colorado State University.
7. Philp, A., 2026: Understanding Hurricane Rapid Intensification Dynamics. RapidWatch technical memorandum, revised 22 June 2026, corrected 13–14 August 2026 — the companion report, source of the ahead-of-track D26 finding cited in Section 06.

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DATA • SHIPS developmental dataset (RAMMB/CIRA, lsdiaga 1982–2023) • HURDAT2 (NHC)  
METHOD • logistic + LDA on PER/SHRD/D200/RHLO/POT/OHC • leave-one-storm-out • Brier Skill Score vs fold climatology  
REPO • aphilp1/gulf-ri-model • commits 9d6acfb (baseline) • 16b6829 (E1) • full tables in results/, write-ups in docs/